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Past Papers Higher Chemistry

2019 Marking Scheme

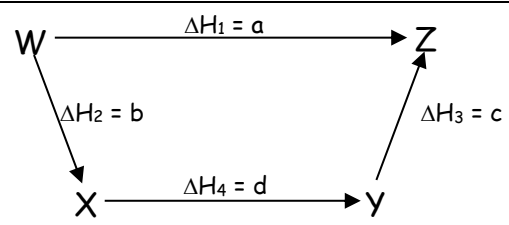
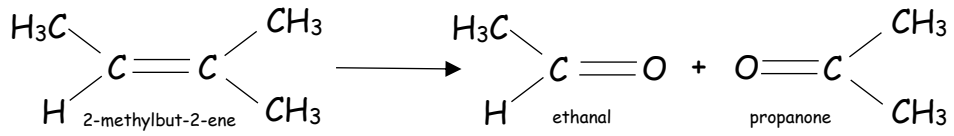
Grade Awarded	Mark Required		% candidates achieving grade
	(/120)	%	
A	80+	66.6%	28.3%
B	67+	55.8%	25.3%
C	54+	45.0%	23.0%
D	47+	39.2%	9.5%
No award	<47	<39.2%	13.9%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	12.1 /20	42.9 /80	13.1 /20

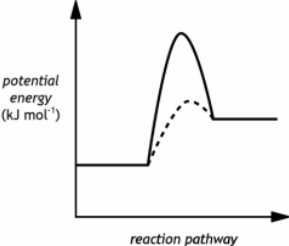
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MC Qu	Answer	% Pupils Correct	Reasoning
1	C	86	Hydrogen has electronegativity of 2.2 so the atom with electronegativity closest to 2.2 would be non-polar when joined to hydrogen.
2	B	76	☒ A CCl_4 is non-polar due to 3D arrangement of Cl atoms around central C atom ☒ B NH_3 is a polar molecule due to the electronegativity difference between atoms ☒ C CO_2 is non-polar due to linear arrangement of O atoms around central C atom ☒ D CH_4 is non-polar as it is a hydrocarbon. All hydrocarbons are non-polar.
3	A	26	☒ A CO reduces metal ores to metals. CO acting as Reducing Agent ☒ B $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ is reduction $\therefore \text{MnO}_4^-$ acting as oxidising agent ☒ C $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}$ is reduction $\therefore \text{H}_2\text{O}_2$ acting as oxidising agent ☒ D $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 5\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ is reduction $\therefore \text{Cr}_2\text{O}_7^{2-}$ acting as oxidising agent
4	C	48	$ \begin{array}{l} \textcircled{1} \quad \text{NO}_3^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{NO} + 2\text{H}_2\text{O} \\ \textcircled{2} \quad \text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^- \\ \textcircled{1} \times 2 \quad 2\text{NO}_3^- + 8\text{H}^+ + \cancel{6\text{e}^-} \rightarrow 2\text{NO} + 4\text{H}_2\text{O} \\ \textcircled{2} \times 3 \quad 3\text{Zn} \rightarrow 3\text{Zn}^{2+} + \cancel{6\text{e}^-} \\ \textcircled{1} + \textcircled{2} \quad 2\text{NO}_3^- + 8\text{H}^+ + 3\text{Zn} \rightarrow 2\text{NO} + 3\text{Zn}^{2+} \\ \qquad \qquad \qquad 2\text{mol} \qquad \qquad \qquad 3\text{mol} \\ \qquad \qquad \qquad 1\text{mol} \qquad \qquad \qquad 1.5\text{mol} \end{array} $
5	B	81	☒ A Primary Alcohol: 1 carbon directly attached to carbon with the -OH group ☒ B Tertiary Alcohol: 3 carbons directly attached to carbon with the -OH group ☒ C Secondary Alcohol: 2 carbons directly attached to carbon with the -OH group ☒ D Secondary Alcohol: 2 carbons directly attached to carbon with the -OH group
6	A	64	☒ A Amino acid with $-\text{CH}_3$ side group has been removed during partial hydrolysis ☒ B $\text{CH}-\text{CH}_3$ group should be on left of $-\text{CH}_2-$ group in hydrolysis fragment ☒ C $-\text{CH}_2-$ group has been removed but two groups are rejoined ☒ D $\text{CH}-\text{CH}(\text{CH}_3)_2$ group should be on right of $-\text{CH}_2-$ group in hydrolysis fragment
7	D	78	☒ A pentyl ethanoate has a total of 7 carbons $\therefore$ cannot have formula $\text{C}_6\text{H}_{12}\text{O}_2$ ☒ B hexan-2-one has formula $\text{C}_6\text{H}_{12}\text{O}$ so only has 1 oxygen atom not 2 oxygen atoms. ☒ C 3-methylpentan-2-ol has formula $\text{C}_6\text{H}_{13}\text{OH}$ so only has 1 oxygen atom not 2. ☒ D hexanoic acid has formula $\text{C}_6\text{H}_{12}\text{O}_2$
8	B	29	Primary Alcohols $\rightarrow$ Hot copper(II) oxide Acidified dichromate solution Aldehydes $\rightarrow$ Hot copper(II) oxide Acidified dichromate solution Carboxylic acids Secondary Alcohol $\rightarrow$ Hot copper(II) oxide Acidified dichromate solution Benedict's or Fehling's solution Tollen's Reagent Ketone ☒ A pentan-1-ol is a primary alcohol would oxidise with hot copper(II) oxide to pentanal which would further oxidise with Fehling's solution to form pentanoic acid ☒ B pentan-2-ol is a secondary alcohol would oxidise with hot copper(II) oxide to form pentan-2-one which does not react with Fehling's solution. ☒ C pentan-3-one is a ketone and does not oxidise with hot copper(II) oxide. ☒ D pentanoic acid is a carboxylic acid and does not oxidise with hot copper(II) oxide.
9	D	88	☒ A molecule has five carbons but not in one continuous chain of five carbons ☒ B longest chain containing functional group has three carbons $\therefore$ ends in propanoic acid ☒ C two separate methyl $-\text{CH}_3$ groups attached to C_2 so is dimethyl not a ethyl $-\text{C}_2\text{H}_5$ group. ☒ D 2x methyl $-\text{CH}_3$ groups attached to C_2 of three carbon main chain with $-\text{COOH}$ group

10	D	72	☒ A Molecule A has 10 carbons and is derived from two C ₅ isoprene units joining together ☒ B Molecule B has 10 carbons and is derived from two C ₅ isoprene units joining together ☒ C Molecule C has 10 carbons and is derived from two C ₅ isoprene units joining together ☒ D Molecule D has 9 carbons so cannot be made by two C ₅ isoprene units joining together.															
11	C	74	☒ A methanol (primary alcohol) oxidises to methanoic acid (carboxylic acid) ☒ B propanal (aldehyde) oxidises to propanoic acid (carboxylic acid) ☒ C butan-2-one (ketone) reduces to become butan-2-ol (secondary alcohol) ☒ D propan-2-ol (secondary alcohol) oxidises to propanone (ketone)															
12	B	92	☒ A Primary Amine: 1 carbon directly bonded to nitrogen atom ☒ B Secondary Amine: 2 carbons directly bonded to nitrogen atom ☒ C Tertiary Amine: 3 carbons directly bonded to nitrogen atom ☒ D Primary Amine: 1 carbon directly bonded to nitrogen atom															
13	D	36	Formula of Calcium Phosphate = Ca ₃ (PO ₄) ₂ 1mol of Ca ₃ (PO ₄) ₂ contains 3mol of Ca ²⁺ ions and 2 mol of PO ₄ ³⁻ ions.															
14	A	65	gfm CH ₄ = 16g ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{4}{16} = 0.25\text{mol}$ ☒ A gfm He = 4g ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{1}{4} = 0.25\text{mol}$ ☒ B gfm H ₂ = 2g ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{1}{2} = 0.5\text{mol}$ ☒ C gfm N ₂ = 28g ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{3.5}{28} = 0.125\text{mol}$ ☒ D gfm Cl ₂ = 71g ∴ no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{35.5}{71} = 0.5\text{mol}$															
15	C	59	$\text{MgCO}_3 + 2\text{HNO}_3 \rightarrow \text{Mg}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$ <table><tr><td>1mol</td><td>2mol</td><td>1mol</td><td>1mol</td><td>1mol</td></tr><tr><td>0.05mol</td><td>0.1mol</td><td>0.05mol</td><td>0.05mol</td><td>0.05mol</td></tr></table> Only 0.06mol of nitric acid available but 0.1mol nitric acid needed to react with all 0.05mol MgCO ₃ ∴ Nitric acid HNO ₃ is the limiting factor. <table><tr><td>0.03mol</td><td>0.06mol</td><td>0.03mol</td><td>0.03mol</td><td>0.03mol</td></tr></table> ☒ A 0.03mol of CO ₂ gas produced ☒ B 0.03mol of MgCO ₃ produced ☒ C 0.03mol of MgCO ₃ reacted ∴ 0.02mol of MgCO ₃ remaining ☒ D nitric acid HNO ₃ is the limiting factor so all 0.06mol are used up.	1mol	2mol	1mol	1mol	1mol	0.05mol	0.1mol	0.05mol	0.05mol	0.05mol	0.03mol	0.06mol	0.03mol	0.03mol	0.03mol
1mol	2mol	1mol	1mol	1mol														
0.05mol	0.1mol	0.05mol	0.05mol	0.05mol														
0.03mol	0.06mol	0.03mol	0.03mol	0.03mol														
16	D	61	☒ A O atom in C=O bonds have δ- charges so will not be attracted to each other ☒ B C-H bond is non-polar due to similar electronegativity so no dipole ☒ C C-H bonds are non-polar due to similar electronegativity so no dipoles ☒ D C in C=O bond has δ+ charge and is attracted to δ- charge on other C=O bond															
17	B	64	atom economy = $\frac{\text{mass of useful products}}{\text{total mass of reactants}} \times 100 = \frac{(4 \times 55.8)}{(2 \times 159.6) + (3 \times 12)} \times 100 = \frac{223.2}{319.2 + 36} \times 100 = 62.8\%$															
18	B	42	$\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$ <table><tr><td>1mol</td><td>5mol</td><td>3mol</td><td>4mol</td></tr><tr><td>1vol</td><td>5vol</td><td>3vol</td><td>negligible volume</td></tr><tr><td>100cm³</td><td>500cm³</td><td>300cm³</td><td>-</td></tr></table> (+100cm ³ O ₂ leftover) Total Volume at end of reaction = 300cm ³ CO ₂ + 100cm ³ leftover O ₂ = 400cm ³	1mol	5mol	3mol	4mol	1vol	5vol	3vol	negligible volume	100cm ³	500cm ³	300cm ³	-			
1mol	5mol	3mol	4mol															
1vol	5vol	3vol	negligible volume															
100cm ³	500cm ³	300cm ³	-															
19	B	45	Step 1: 60% of 100% = $\frac{60}{100} \times 100\% = 60\%$ Step 2: 90% of 60% = $\frac{90}{100} \times 60\% = 54\%$															
20	C	50	☒ A Volume of gas must be reduced as volume of acid is reduced (zinc in excess) ☒ B Initial Rate of reaction must be increased as lumps replaced by powder ☒ C Initial rate must be greater and final volume of gas must be reduced ☒ D Initial Rate of reaction must be increased as lumps replaced by powder															

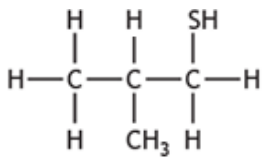
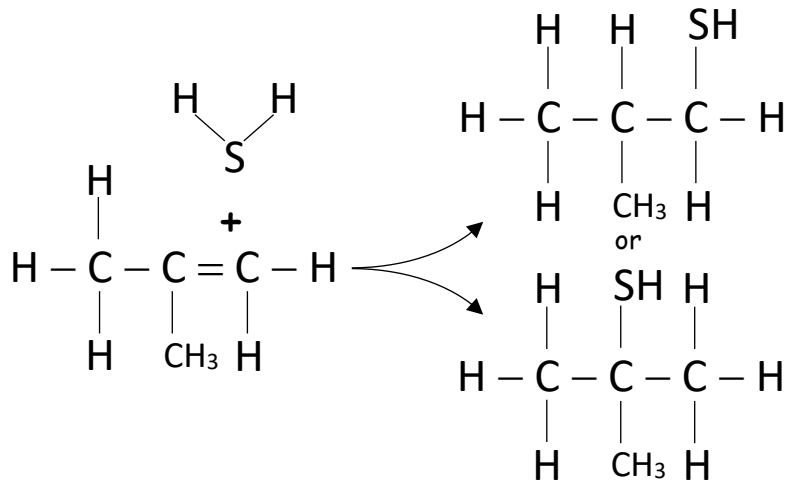
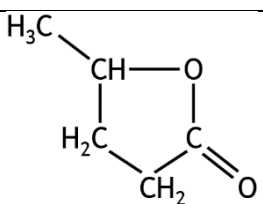
21	A	70	$\Delta H_1 = \Delta H_2 + \Delta H_3 + \Delta H_4$ $\Delta H_2 = \Delta H_1 - \Delta H_3 - \Delta H_4$ $b = a - c - d$ 
22	D	85	☒ A Higher Activation Energy will make a successful collision less likely to happen. ☒ B The higher the kinetic energy of reactants the more like the collision will have sufficient energy to react. ☒ C Higher the concentration the higher the likelihood of a successful collision ☒ D Whether a reaction is exothermic or endothermic has no bearing on the reaction rate.
23	C	71	☒ A no change in pressure from reactants to products ∴ lowering pressure has no effect ☒ B lowering pressure favours pressure increasing reaction (reverse reaction) ☒ C lowering pressure favours pressure increasing reaction (forward reaction) ☒ D lowering pressure favours pressure increasing reaction (reverse reaction)
24	A	71	☒ A Increasing the temperature moves the curve to the right. ☒ B Increasing the temperature moves the curve to the right not the left. ☒ C E_a does not change when temperature is changed ☒ D Area under curve should be same as same number of particles.
25	D	47	

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Long Qu	Answer	Reasoning										
1a(i)	Answer showing:	$\text{Na}_2\text{S}_2\text{O}_3 + 2\text{HCl} \longrightarrow \text{S} + \text{SO}_2 + 2\text{NaCl} + \text{H}_2\text{O}$										
1a(ii) Part A	<table border="1"><tr><td>A</td><td>0</td></tr><tr><td>B</td><td>10</td></tr><tr><td>C</td><td>20</td></tr><tr><td>D</td><td>30</td></tr><tr><td>E</td><td>40</td></tr></table>	A	0	B	10	C	20	D	30	E	40	For the concentration of thiosulphate to be varied, the total volume of the solution must be kept constant. The total volume of sodium thiosulphate solution and water is 50 cm ³ in each experiment.
A	0											
B	10											
C	20											
D	30											
E	40											
1a(ii) Part B	35.1	$\text{Rate} = \frac{1}{\text{Time}} \quad \therefore \text{time} = \frac{1}{\text{Rate}} = \frac{1}{0.0285} = 35.1\text{s}$										
1a(iii)	12±1	For doubling of rate from 0.02s⁻¹ to 0.04s⁻¹ Temperature at 0.02s⁻¹ = 44°C Temperature at 0.04s⁻¹ = 56°C } Change in temperature = 12°C										
1b	Sufficient Energy to React And Correct Geometry	<table border="1"><tr><td>1st Mark:</td><td>sufficient or enough energy</td><td>energy equal to or greater than the activation energy</td><td>minimum/ enough energy to form an activated complex</td></tr><tr><td>2nd Mark:</td><td colspan="3">(Collision must occur with) suitable/correct/geometry/orientation</td></tr></table>	1 st Mark:	sufficient or enough energy	energy equal to or greater than the activation energy	minimum/ enough energy to form an activated complex	2 nd Mark:	(Collision must occur with) suitable/correct/geometry/orientation				
1 st Mark:	sufficient or enough energy	energy equal to or greater than the activation energy	minimum/ enough energy to form an activated complex									
2 nd Mark:	(Collision must occur with) suitable/correct/geometry/orientation											
1c(i)	X at peak on curve	The top of the hill (peak on the curve) is the activated complex where the bonds of the reactants are half broken and the bonds of the products are half formed.										
1c(ii)		A catalyst lowers the activation energy without changing the position of the reactants or products. This means that the top of the hill is lowered. The enthalpy change is the same as the positions of the reactants and products are unchanged.										
2a(i)	Increasing number of protons or increasing nuclear charge	Going across a period does not increase the size of an atom as it is the same outer shell which is being filled up. The increased positive charge in the nucleus attracts the outer shell into more as you go across a period.										
2a(ii)	One answer from:	<table border="1"><tr><td>Increased screening/shielding</td><td>Covalent radius increases</td><td>Atom size increases</td><td>More shells so less attraction of nucleus for outer electron decreases</td></tr></table>	Increased screening/shielding	Covalent radius increases	Atom size increases	More shells so less attraction of nucleus for outer electron decreases						
Increased screening/shielding	Covalent radius increases	Atom size increases	More shells so less attraction of nucleus for outer electron decreases									
2b(i)	$\text{N}^+(\text{g}) \rightarrow \text{N}^{2+}(\text{g}) + \text{e}^-$	1 st Ionisation Energy: The removal of one mole of electrons from one mole of atoms in the gaseous state. 2 nd Ionisation Energy: The removal of one mole of electrons from one mole of 1+ ions in the gaseous state.										
2b(ii)	Answer to Include:	<table border="1"><tr><td>1st Mark:</td><td>The 6th ionisation energy involves removing an electron from the shell which is full/stable/closer to the nucleus</td><td>or</td><td>the 6th electron is removed from the electron shell which is inner/full/stable/closer to the nucleus</td></tr><tr><td>2nd Mark:</td><td>The 6th electron is less shielded</td><td>or</td><td>the 6th electron is more strongly attracted to/pulled towards the nucleus.</td></tr></table>	1 st Mark:	The 6th ionisation energy involves removing an electron from the shell which is full/stable/closer to the nucleus	or	the 6th electron is removed from the electron shell which is inner/full/stable/closer to the nucleus	2 nd Mark:	The 6th electron is less shielded	or	the 6th electron is more strongly attracted to/pulled towards the nucleus.		
1 st Mark:	The 6th ionisation energy involves removing an electron from the shell which is full/stable/closer to the nucleus	or	the 6th electron is removed from the electron shell which is inner/full/stable/closer to the nucleus									
2 nd Mark:	The 6th electron is less shielded	or	the 6th electron is more strongly attracted to/pulled towards the nucleus.									
2c	1st Mark: Al forms Al³⁺ ion P forms P³⁻ ion 2nd Mark: P³⁻ ion has one more electron shell than Al³⁺ ion	Phosphorus atoms have electron arrangement of 2,8,5 and form P³⁻ ions which have electron arrangement of 2,8,8 Aluminium atoms have electron arrangement of 2,8,3 and form Al³⁺ ions which have electron arrangement of 2,8 Phosphide P³⁻ ion has one more electron shell than aluminium Al³⁺ ion.										
2d	Radius Ratio = 0.96 Caesium Chloride Structure	$\text{Radius ratio} = \frac{\text{Radius of positive ion}}{\text{Radius of negative ion}} = \frac{135}{140} = 0.96$										

3	Open Question Answer to Include:	3 mark answer	2 mark answer	1 mark answer	
		Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.	
4a(i)	Biological catalyst	An enzyme is a protein which acts as a catalyst for the chemical reactions inside living things.			
4a(ii)	4.5	Mass of cider = $1.36\text{g}/\text{cm}^3 \times 50\text{cm}^3 = 68\text{g}$ $\% \text{mass of alcohol} = \frac{\text{Mass of alcohol}}{\text{Mass of cider}} \times 100 = \frac{3.05}{68} \times 100 = 4.48\%$			
4b(i)	Carbon dioxide	$\begin{array}{ccccccc} \text{Malic acid} & \longrightarrow & \text{Lactic Acid} & + & \text{X} \\ \text{C}_4\text{H}_6\text{O}_5 & \longrightarrow & \text{C}_3\text{H}_6\text{O}_3 & + & \text{CO}_2 \end{array}$			
4b(ii) Part A	0.25	$R_f = \frac{\text{Distance moved by substance}}{\text{Distance moved by solvent}} = \frac{4.1}{16.4} = 0.25$			
4b(ii) Part B	Sample 4 or Cider B	Problem Solving: Cider B/Sample 4 is the only cider that has no malic acid spot at 4.1 indicating all the malic acid has turned into lactic acid			
4c	Propane-1,2,3-triol	$\begin{array}{ccccc} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & \\ \text{OH} & \text{OH} & \text{OH} \\ \text{glycerol} \end{array}$ Propane-1,2,3-triol 3 carbons Single bonds between carbons Functional groups on Carbons 1,2,3 3 hydroxyl -OH groups			
4d(i)	Answer to include:	1 st Mark:	Molecule is polar due to hydroxyl groups or Can form hydrogen bonds due to hydroxyl groups		
		2 nd Mark:	Solubility increases as more polar hydroxyl groups are added (and increases hydrogen bonding)		
4d(ii)	2-methylbutanoic acid	$\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{O} & & \text{H} & \text{H} & \text{H} & \text{O} \\ & & & & & & & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{C}-\text{C}-\text{H} & \longrightarrow & \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ & & & & & & & & \\ \text{H} & \text{H} & \text{CH}_3 & & & \text{H} & \text{H} & \text{CH}_3 & \\ \text{ethyl 2-methylbutanoate} & & & & & \text{2-methylbutanoic acid} \end{array}$ + H₂O $\begin{array}{ccc} \text{H} & \text{H} \\ & \\ \text{H}-\text{O}-\text{C}-\text{C}-\text{H} \\ & \\ \text{H} & \text{H} \\ \text{ethanol} \end{array}$			
4d(iii)	2-methylbuta-1,3-diene or isoprene	$\begin{array}{c} \text{H}_2\text{C}^{\textcircled{1}} \\ \parallel \\ \text{C}^{\textcircled{2}}-\text{CH}^{\textcircled{3}} \\ \qquad \parallel \\ \text{H}_3\text{C} \qquad \text{CH}_2^{\textcircled{4}} \end{array}$ 2-methylbuta-1,3-diene			
4e(i)	Carbonyl	$\begin{array}{c} \text{O} \\ \\ -\text{C}- \end{array}$ carbonyl group	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$ carboxyl group	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{H} \end{array}$ aldehyde group	$\begin{array}{c} \text{O} \\ \\ \text{C}-\text{C}-\text{C} \end{array}$ ketone group

4e(ii)	Ethanoic acid	Primary Alcohol $\longrightarrow$ Aldehyde $\longrightarrow$ Carboxylic acid Ethanol $\longrightarrow$ Ethanal $\longrightarrow$ Ethanoic acid																
5a(i)	-694	<table><tr><th colspan="2">Bond Breaking Steps</th><th colspan="2">Bond Forming Steps</th></tr><tr><td>4x C-H bonds</td><td>4x 412kJ = 1648kJ</td><td>2x C=O bonds</td><td>2x 743kJ = 1486kJ</td></tr><tr><td>2x O=O bond</td><td>2x 498kJ = 996kJ</td><td>4x O-H bonds</td><td>4x 463kJ = 1852kJ</td></tr><tr><td>Total bond breaking</td><td>= 2644kJ</td><td>Total bond Forming</td><td>= 3338kJ</td></tr></table> Enthalpy change = +2644 - 3338 = -694kJ mol⁻¹ $\Delta H = \Sigma \text{Bond enthalpies for bonds broken} - \Sigma \text{Bond enthalpies for bonds formed}$ $\Delta H = 2644 - 3338$ $\Delta H = -694 \text{ kJ mol}^{-1}$	Bond Breaking Steps		Bond Forming Steps		4x C-H bonds	4x 412kJ = 1648kJ	2x C=O bonds	2x 743kJ = 1486kJ	2x O=O bond	2x 498kJ = 996kJ	4x O-H bonds	4x 463kJ = 1852kJ	Total bond breaking	= 2644kJ	Total bond Forming	= 3338kJ
Bond Breaking Steps		Bond Forming Steps																
4x C-H bonds	4x 412kJ = 1648kJ	2x C=O bonds	2x 743kJ = 1486kJ															
2x O=O bond	2x 498kJ = 996kJ	4x O-H bonds	4x 463kJ = 1852kJ															
Total bond breaking	= 2644kJ	Total bond Forming	= 3338kJ															
5a(ii)	Answer to include:	Mean bond enthalpy is an average energy from a number of compounds. Bond enthalpy relates to only one particular compound or molecule.																
5a(iii)	0.367	no. of moles = $\frac{\text{Volume}}{\text{Molar Volume}} = \frac{0.200\text{litres}}{24 \text{ litres mol}^{-1}} = 0.00833\text{mol}$ $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ 1mol 2mol 1mol 2mol 0.00833mol 0.00833mol mass = no. of mol x gfm = 0.00833 x 44 = 0.367g																
5b(i)	Record the mass of burner before and after heating	The before and after masses of the spirit burner (including lid) are needed to calculate the change in mass of the spirit burner and this change in mass is the mass of heptane burned.																
5b(ii)	-3496	Heat Energy = Specific Heat Capacity x Mass x Change In Temperature $E_h = c \times m \times \Delta T$ $E_h = 4.18 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1} \times 0.4\text{kg} \times 23^\circ\text{C}$ $E_h = 38.456 \text{ kJ}$ gfm Heptane C₇H₁₆ = (7x12) + (16x1) = 84 + 16 = 100g 1.1g heptane $\longleftrightarrow$ 38.456kJ 1mol heptane = 100g heptane $\longleftrightarrow$ 38.456kJ x ¹⁰⁰/_{1.1} = -3496kJ mol⁻¹																
5b(iii)	One answer from:	<table><tr><td>Loss of heat to surroundings</td><td>Incomplete combustion</td><td>Loss by evaporation</td></tr><tr><td>Absorption of heat by glass/beaker/can</td><td>No stirring</td><td>No lid on container</td></tr></table>	Loss of heat to surroundings	Incomplete combustion	Loss by evaporation	Absorption of heat by glass/beaker/can	No stirring	No lid on container										
Loss of heat to surroundings	Incomplete combustion	Loss by evaporation																
Absorption of heat by glass/beaker/can	No stirring	No lid on container																
6a(i) Part A	Same number of electrons or Same strength of LDF	Both molecules have identical number of electrons (34) so have the same ability to form London Dispersion Forces between molecules (due to formation of temporary dipoles in their electron clouds). The difference between the molecules must be caused by other intermolecular forces.																
6a(i) Part B	Answer to include:	<table><tr><td>1st mark:</td><td>Propan-1-ol has stronger intermolecular forces than ethanethiol</td><td>Intermolecular forces in propan-1-ol take more energy to break than those in ethanethiol</td></tr><tr><td>2nd mark:</td><td colspan="2">Intermolecular bonds in propan-1-ol are hydrogen bonds and intermolecular bonds in ethanethiol are permanent dipole to permanent dipole attractions</td></tr></table>	1 st mark:	Propan-1-ol has stronger intermolecular forces than ethanethiol	Intermolecular forces in propan-1-ol take more energy to break than those in ethanethiol	2 nd mark:	Intermolecular bonds in propan-1-ol are hydrogen bonds and intermolecular bonds in ethanethiol are permanent dipole to permanent dipole attractions											
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6a(ii)	methanethiol	<table><tr><td>Alkane:</td><td>Methane CH₄</td><td>Ethane C₂H₆</td><td>Propane C₃H₈</td></tr><tr><td>Thiol:</td><td>Mathanethiol CH₃SH</td><td>Ethanethiol C₂H₅SH</td><td>Propanethiol C₃H₇SH</td></tr></table>	Alkane:	Methane CH ₄	Ethane C ₂ H ₆	Propane C ₃ H ₈	Thiol:	Mathanethiol CH ₃ SH	Ethanethiol C ₂ H ₅ SH	Propanethiol C ₃ H ₇ SH								
Alkane:	Methane CH ₄	Ethane C ₂ H ₆	Propane C ₃ H ₈															
Thiol:	Mathanethiol CH ₃ SH	Ethanethiol C ₂ H ₅ SH	Propanethiol C ₃ H ₇ SH															
6a(iii)	11.853mg	1cm³ air $\longleftrightarrow$ 2.7x10⁻⁷mg 1 litre air $\longleftrightarrow$ 2.7x10⁻⁴mg 43900 litres air $\longleftrightarrow$ 2.7x10⁻⁴mg x ⁴³⁹⁰⁰/₁ = 11.853mg or 0.0118g																
6b(i)	-SH group is attached to carbon which is attached to 3 other carbons	<table><tr><th>Primary Thiol</th><th>Secondary Thiol</th><th>Tertiary Thiol</th></tr><tr><td>-SH group attached to carbon which is attached to 0 or 1 other carbons atoms</td><td>-SH group attached to carbon which is attached to 2 other carbon atoms.</td><td>-SH group attached to carbon which is attached to 3 other carbon atoms.</td></tr><tr><td>-SH group attached to carbon which is attached to 2 hydrogen atoms.</td><td>-SH group attached to carbon which is attached to 1 hydrogen atom.</td><td>-SH group attached to carbon which is attached to no hydrogen atoms.</td></tr></table>	Primary Thiol	Secondary Thiol	Tertiary Thiol	-SH group attached to carbon which is attached to 0 or 1 other carbons atoms	-SH group attached to carbon which is attached to 2 other carbon atoms.	-SH group attached to carbon which is attached to 3 other carbon atoms.	-SH group attached to carbon which is attached to 2 hydrogen atoms.	-SH group attached to carbon which is attached to 1 hydrogen atom.	-SH group attached to carbon which is attached to no hydrogen atoms.							
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6b(ii) Part A														
6b(ii) Part B	41.2	2-methylpropene + hydrogen sulphide $\longrightarrow$ 2-methyl-2-propanethiol 1mol 56.0g 30.5g 1mol 90.1g $90.1g \times \frac{30.5}{56.0}$ $= 49.07g$ $\% \text{Yield} = \frac{\text{Actual}}{\text{Theoretical}} \times 100 \quad \therefore \text{Actual} = \frac{\% \text{Yield} \times \text{Theoretical}}{100} = \frac{84 \times 49.07}{100} = 41.2g$												
7a(i)	Propagation	<table border="1"> <thead> <tr> <th>Step</th><th>Reactants (before Arrow)</th><th>Products (after Arrow)</th></tr> </thead> <tbody> <tr> <td>Initiation</td><td>No free radicals on Reactant Side</td><td>Free radicals on Product Side</td></tr> <tr> <td>Propagation</td><td colspan="2">Free Radicals found on both sides of arrow</td></tr> <tr> <td>Termination</td><td>Free radicals on Reactant Side</td><td>No free radicals on Product Side</td></tr> </tbody> </table>	Step	Reactants (before Arrow)	Products (after Arrow)	Initiation	No free radicals on Reactant Side	Free radicals on Product Side	Propagation	Free Radicals found on both sides of arrow		Termination	Free radicals on Reactant Side	No free radicals on Product Side
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Termination	Free radicals on Reactant Side	No free radicals on Product Side												
7a(ii)	uv/ultraviolet	Ultraviolet light can cause the formation of free radicals as energy in the uv light can cause bonds to split and the two electrons in the bond separate one to each side. This means there are unpaired electrons which are called free radical. This breaks the plastic down in to smaller chunks that can be digested by bacteria.												
7a(iii)	Anti-oxidant or Free Radical Scavenger	Free Radical Scavengers and anti-oxidants quickly react with any free radical particles going and prevent future propagation steps which would prolong the breakdown of the plastics.												
7b(i)	Water/H ₂ O	5-hydroxypentanoic acid = C ₅ H ₁₀ O ₃ lactone = C ₅ H ₈ O ₂ Difference = H ₂ O												
7b(ii)		One less carbon between Carboxyl -COOH group and Hydroxyl group $\therefore$ One less carbon in lactone ring i.e. ring has 4 carbons plus 1 oxygen in ring Carbon with hydroxyl -OH group has methyl -CH₃ group sticking off it $\therefore$ Methyl -CH₃ group sticking off C on other side of -O-C=O ester group												
7b(iii)	3-hydroxybutanoic acid	3-hydroxybutanoic acid -OH on C₃ -OH side group Four carbon main chain Carboxyl -COOH functional group												

8a(i)	6	amino acid 1 amino acid 2 amino acid 3 amino acid 4 amino acid 5 amino acid 6																																																								
8a(ii)	London Dispersion Forces	There are three forms of van der Waals' Attraction. London Dispersion forces are found in all substances but are the weakest form of intermolecular attraction. Permanent dipole to permanent dipole attractions are stronger than London Dispersion Forces and Hydrogen Bonding is the strongest form of van der Waals' attraction.																																																								
8b(i)	Answer to include:	<table><tr><td>1st Mark:</td><td>Dissolve gelatin (in small volume of deionised water)</td></tr><tr><td>2nd Mark:</td><td>Transfer quantitatively/with rinsings/with washings</td></tr><tr><td>3rd Mark:</td><td>Fill to the mark/line (on volumetric/standard flask)</td></tr></table>	1 st Mark:	Dissolve gelatin (in small volume of deionised water)	2 nd Mark:	Transfer quantitatively/with rinsings/with washings	3 rd Mark:	Fill to the mark/line (on volumetric/standard flask)																																																		
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8b(ii)	11.0	<table><tr><td>Concentration (%)</td><td>2.0</td><td>4.0</td><td>6.0</td><td>8.0</td><td>10.0</td></tr><tr><td>Viscosity (units)</td><td>1.0</td><td>2.0</td><td>4.0</td><td>7.0</td><td>(-)</td></tr><tr><td>Difference</td><td></td><td>1.0</td><td>2.0</td><td>3.0</td><td>(4.0)</td></tr><tr><td>Prediction (units)</td><td>(-)</td><td>(-)</td><td>(-)</td><td>(-)</td><td>11.0</td></tr></table>	Concentration (%)	2.0	4.0	6.0	8.0	10.0	Viscosity (units)	1.0	2.0	4.0	7.0	(-)	Difference		1.0	2.0	3.0	(4.0)	Prediction (units)	(-)	(-)	(-)	(-)	11.0																																
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Prediction (units)	(-)	(-)	(-)	(-)	11.0																																																					
8c(i)	Enzyme Changes shape or denatured	Enzymes are specifically shaped globular proteins which denature when heated. Denaturing is caused by the 3D structure of the protein in the enzyme changing. This 3D structure is held by various types of bonding e.g. hydrogen bonding. Once the enzyme has changed shape, the substrate molecule no longer fits the enzyme active site and the enzyme no longer catalyses the reaction.																																																								
8c(ii)	37.88	13.2mg bromelain ↔ 1g pineapple 500mg bromelain ↔ 1g × ⁵⁰⁰/_{13.2} = 37.88g																																																								
9a(i)	+220±2	Activation Energy (forward reaction) is measure from: R to Activated Complex = 220 - 0 = +220kJ mol ⁻¹ NB: Activation energy are always endothermic with a positive value.																																																								
9a(ii)	One Answer from:	<table><tr><td>Favours the endothermic/ reverse reaction</td><td>(Forward) reaction is exothermic</td><td>Reverse reaction is endothermic</td></tr></table>	Favours the endothermic/ reverse reaction	(Forward) reaction is exothermic	Reverse reaction is endothermic																																																					
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9b	Diagram showing:	<table><tr><td>workable method for removal of HCl but allowing Cl₂ to pass through water (1mark)</td><td>workable method to collect gas (1mark)</td></tr><tr><td></td><td></td></tr></table>	workable method for removal of HCl but allowing Cl ₂ to pass through water (1mark)	workable method to collect gas (1mark)																																																						
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9c	-391	<table><tr><td>①</td><td>C</td><td>+</td><td>2H₂</td><td>→</td><td>CH₄</td><td>ΔH=-75 kJ mol⁻¹</td></tr><tr><td>②</td><td>C</td><td>+</td><td>2Cl₂</td><td>→</td><td>CCl₄</td><td>ΔH=-98 kJ mol⁻¹</td></tr><tr><td>③</td><td>$\frac{1}{2}\text{H}_2$</td><td>+</td><td>$\frac{1}{2}\text{Cl}_2$</td><td>→</td><td>HCl</td><td>ΔH=-92 kJ mol⁻¹</td></tr><tr><td>①x-1</td><td></td><td></td><td>CH₄</td><td>→</td><td>C + 2H₂</td><td>ΔH=+75 kJ mol⁻¹</td></tr><tr><td>②</td><td>C</td><td>+</td><td>2Cl₂</td><td>→</td><td>CCl₄</td><td>ΔH=-98 kJ mol⁻¹</td></tr><tr><td>③x4</td><td>2H₂</td><td>+</td><td>2Cl₂</td><td>→</td><td>4HCl</td><td>ΔH=-368 kJ mol⁻¹</td></tr><tr><td>Add</td><td colspan="6"></td></tr><tr><td>①'+②+③'</td><td>CH₄</td><td>+</td><td>4Cl₂</td><td>→</td><td>CCl₄ + 4HCl</td><td>ΔH=-391 kJ mol⁻¹</td></tr></table>	①	C	+	2H ₂	→	CH ₄	ΔH=-75 kJ mol ⁻¹	②	C	+	2Cl ₂	→	CCl ₄	ΔH=-98 kJ mol ⁻¹	③	$\frac{1}{2}\text{H}_2$	+	$\frac{1}{2}\text{Cl}_2$	→	HCl	ΔH=-92 kJ mol ⁻¹	①x-1			CH ₄	→	C + 2H ₂	ΔH=+75 kJ mol ⁻¹	②	C	+	2Cl ₂	→	CCl ₄	ΔH=-98 kJ mol ⁻¹	③x4	2H ₂	+	2Cl ₂	→	4HCl	ΔH=-368 kJ mol ⁻¹	Add							①'+②+③'	CH ₄	+	4Cl ₂	→	CCl ₄ + 4HCl	ΔH=-391 kJ mol ⁻¹
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12d(ii)	1.76×10^{-2} or 0.0176	no. of mol = volume $\times$ concentration = $0.0090 \text{ litres} \times 0.098 \text{ mol l}^{-1} = 8.82 \times 10^{-4} \text{ mol}$ $ \begin{array}{ccccccc} \text{I}_2 & + & 2\text{Na}_2\text{S}_2\text{O}_3 & \longrightarrow & 2\text{NaI} & + & \text{Na}_2\text{S}_2\text{O}_6 \\ 1\text{mol} & & 2\text{mol} & & & & \\ 4.41 \times 10^{-4} \text{ mol} & & 8.82 \times 10^{-4} \text{ mol} & & & & \end{array} $ $ \begin{array}{ccccccc} \text{OCl}^- & + & 2\text{I}^- & + & 2\text{H}^+ & \longrightarrow & \text{I}_2 + \text{Cl}^- + \text{H}_2\text{O} \\ 1\text{mol} & & & & & & 1\text{mol} \\ 4.41 \times 10^{-4} \text{ mol} & & & & & & 4.41 \times 10^{-4} \text{ mol} \end{array} $ concentration = $\frac{\text{no. of mol}}{\text{volume}} = \frac{4.41 \times 10^{-4} \text{ mol}}{0.025 \text{ litres}} = 1.76 \times 10^{-2} \text{ mol l}^{-1}$
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2019 Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	C	4	2	86	8	0	0
2	B	4	76	15	3	0	1
3	A	26	23	20	28	0	2
4	C	15	24	48	12	1	1
5	B	9	81	7	3	0	1
6	A	64	7	22	7	0	1
7	D	8	8	4	78	0	1
8	B	6	29	53	11	0	1
9	D	1	10	1	88	0	0
10	D	5	2	22	72	0	1
11	C	6	8	74	11	0	1
12	B	5	92	2	1	0	1
13	D	24	21	18	36	0	1
14	A	65	6	26	2	1	0
15	C	12	7	59	21	1	1
16	D	10	19	9	61	0	1
17	B	2	64	6	27	0	1
18	B	32	42	15	10	1	1
19	B	12	45	18	23	1	1
20	C	16	28	50	5	0	1
21	A	70	2	4	24	0	1
22	D	10	3	1	85	0	1
23	C	6	14	71	7	0	1
24	A	71	6	17	5	0	1
25	D	39	5	8	47	0	1